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Data Augmented Neural Network for Temporal Segmentation of Electromyography Signals

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Abstract

In this paper, we investigated how temporal deep learning models can be improved for segmentation and classification of Electromyography (EMG) signals for hand / arm gesture recognition. We used the EMG-EPN-612 dataset and developed a baseline Temporal Convolutional Network (TCN) and expanded it with a data augmentation that integrates two manually calculated temporal features: Mean Absolute Value (MAV) and waveform length (WL). These features are computed directly from the raw 8-channel EMG recordings and are designed to provide the network with additional descriptive indicators of muscle activity within each windowed segment of the signal. We also implemented a sliding-window and window-overlap approach within windowed intervals. This enabled the model to preserve short range temporal context and maintained a consistent structure across samples. Our results show that the augmented TCN achieves an average improvement of 9.13% in correct predictions across all gesture classes, demonstrating that manual data augmentation from raw source can improve temporal pattern recognition in EMG-based systems. The study highlights both the potential of TCNs in realtime prosthetic control and the need for future work in feature design, window alignment, and cross-user generalization to further improve performance in different domains and applications.

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Introduction

In a world where technology is rapidly advancing, many focus on aspects that revolve around adapting such technology to improve itself over time. However, in this paper, I want to extrapolate this concept and instead focus on the integration of this rapid advancement within the medical field, specifically in prosthetics for human extremities. There are several domains that we can examine; in this document, we will cover Electromyography signals, how they are captured, processed, and integrated within prosthetics. My main objective is to articulate the concept of integrating Machine Learning with this domain, extend the reader's understanding of how powerful such tools can be in improving quality of life, and highlight how Deep Learning can further revolutionize this area.

Let us begin by breaking down what EMG signals are. EMG stands for Electromyography. These signals are captured either by inserting a small needle into a muscle to record electrical differences or by using skin-surface electrodes that conduct these readings without any subcutaneous injections (Farina et al., 2016; Chowdhury et al., 2013; Reaz et al., 2006). These probes measure electrical differences as the muscle contracts during specific movements. For example, if I wanted to measure which muscles activate and how they behave when I make a fist or rotate my wrist, I could place this device around my arm and measure, as a time series, the electrical impulses of each muscle group. This represents a simple subset of how EMG signals are utilized. In this research document, we will focus specifically on EMGs, emphasizing datasets that capture wrist and arm activity based on hand gestures. This allows us to narrow the scope of the study and reduce the need for extensive knowledge of more complex muscle groups.

Anatomy of EMG Signals & Cross User Generalization

Now that we understand what EMG signals are at a high level, let us examine an example of output data from a given test subject. The Figure 1 below shows an action performed by a participant in this case, flexing the hand and wrist. The signal graph to the left displays a composite reading of every signal measured around the wrist. In this example, the device used is an 8-channel Myo Armband, which measures muscle contractions circumferentially around the forearm.

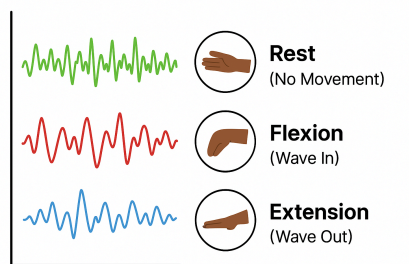


Figure [1]: An AI-generated visual representation of gestures via EMG readings.

One critical aspect to understand about this signal is that it captures muscle contraction activity from the entire wrist in a 360-degree manner, not solely from a single area. In our case, we will later break down these signals and reconstruct them to gain a better understanding of how these contractions evolve over time and how they can be segmented into discrete time-series windows. This methodology can naturally be extended to different extremities to achieve similar outcomes for a variety of research objectives.

Although the data is represented numerically via the output signals to achieve a standardized modality, the primary concern is cross-user variability for specific extremities. It is difficult to maintain consistent wrist size, mobility efficiency, and muscle strength across a large pool of test subjects. This issue can be mitigated by expanding the dataset to include a broader demographic

or by focusing on a more homogeneous user group. For the purposes of this project, we will utilize the first 200 users provided from the dataset.

EMG-EPN-612 Data Source

The data utilized for this research document is sourced from LibEMG, specifically the EMG-EPN-612 dataset (Benalcázar et al., 2021; Laboratorio de Investigación en Inteligencia y Visión Artificial, 2020) which contains recordings from 612 users. Each user performs six distinct gestures, repeated 50 times, targeting hand and wrist movements. The device in question is an 8-channel Myo Armband that records EMG activity at 200 Hz. This dataset is sufficient for the objectives of our experiment and provides clearly separated training and testing subsets.

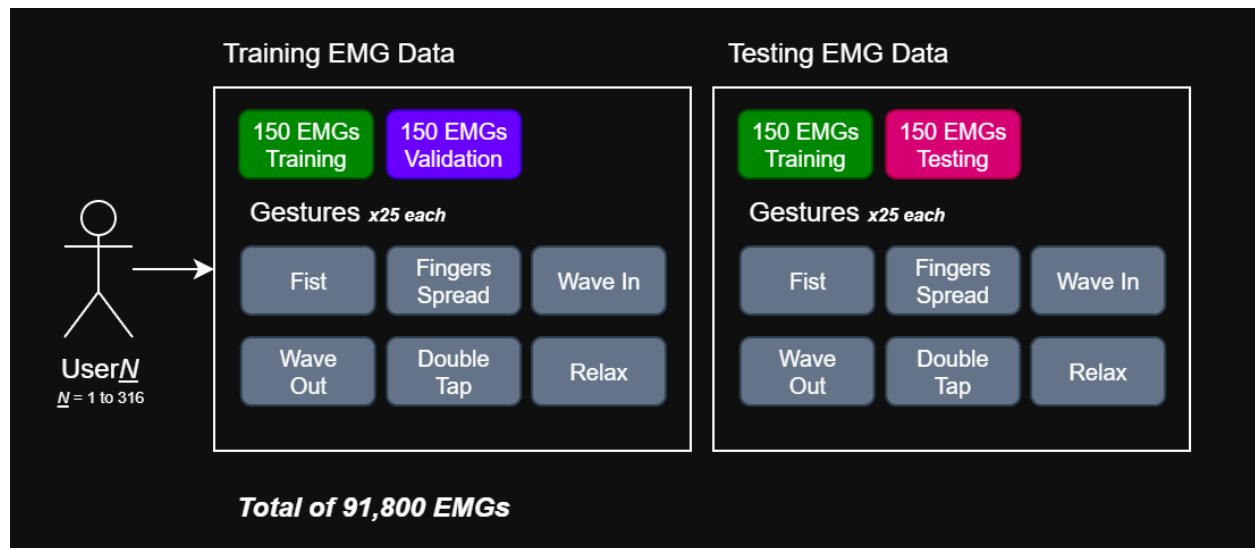


Figure [2]: How EMG-EPN-612 data is represented.

Figure 2 above illustrates a simplified breakdown of the data captured for each gesture. The dataset is accessed by interfacing with either the training or testing folders, each containing user-specific entries. Each user object includes a list of 600 snapshots corresponding to the

recorded gestures. These snapshots can be interpreted as individual instances of the Myo Armband EMG readings each spanning 5 seconds of recorded EGM activity.

Background and Prior Work

Signal processing in general has a long history in Machine Learning, but the specific applications for prosthetics and robotics have only accelerated in recent decades. There is a lot of work done for our chosen dataset, especially from the research paper CNN-LSTM and post-processing for EMG-based hand gesture recognition (Lopez et al., 2024, p. 7), which presents an excellent Convolutional Neural Network with Long Short-Term Memory. In their study, López et al. used spectrogram-based inputs and showed that their CNN-LSTM model reached 92.0% classification accuracy and 90.55% recognition accuracy, largely due to a post-processing step that improved performance by 24.77%, even more than the LSTM itself. Their findings highlight how temporal modeling and careful filtering of predictions are key when working with noisy, multi-user EMG data. In our research paper, we move away from LSTM and use a general one dimensional convolutional layer to mimic the Temporal nature of model learning. Recall that LSTM is a specific model architecture under the general term of Temporal Convolutional Networks, just like Deep Learning is a subset of Machine Learning.

Research Scope

As time-series data becomes increasingly prominent in domains centered on signal processing, expanding beyond traditional convolutional neural networks and incorporating Temporal Convolutional Networks (TCNs) presents a promising direction. Current research surrounding EMG signals emphasizes minimizing the window length required for a high-accuracy model to predict a given motion (Valdivieso Caraguay et al., 2023; Shankar &

Tamilarasi, 2025), effectively requiring models to infer more with less data. This is one of the primary reasons I chose TCNs: they retain contextual information across segmented windows of signal data functionally similar to a moving average that reflects how the data behaves over the preceding 100 milliseconds.

In this paper, I will examine existing methodologies that utilize such networks and develop a baseline model trained on the previously described dataset. The ultimate objective is to investigate how data augmentation influences TCN performance and whether these enhancements improve the network's ability to recognize patterns within EMG signals.

Research Methods

Let us begin by defining the main differences between TCNs and CNNs. TCNs, or Temporal Convolutional Neural Networks, are convolution-based deep learning models designed specifically for sequential data (a data source composed of continuous values, as seen in signal processing) (Lea et al., 2016). These networks are particularly powerful when the underlying signal depends on knowledge or pattern recognition derived from previous observations within a short temporal window. Since our data is explicitly time-dependent, capturing muscle contractions across a continuous time series, TCNs ensure that causality is preserved from the onset of a contraction through the remainder of the signal (e.g. from time t to $t + 30\text{ms}$).

The fundamental distinction between TCNs and CNNs lies in the nature of the data they are intended to model. TCNs operate on one-dimensional temporal sequences, while CNNs are traditionally used for two-dimensional spatial feature extraction such as image classification,

object detection, and similar tasks (Ajit et al., 2020; Li et al., 2021; Sze et al., 2017). Another important difference is temporal awareness: TCNs capture temporal order and directionality, whereas CNNs are not inherently designed to retain historical information. In summary, CNNs are robust in extracting spatial features from multidimensional data but lack temporal focus. In contrast, TCNs embed memory, causality, and temporal context, all of which directly align with the objectives of our model.

Temporal Focus and Window Segmentation

A unique feature of TCNs is their ability to retain information across high variability in cross-user data sources. This enables the network to recognize specific patterns within each unique gesture and learn them directly from the raw stream of EMG data. In our model, we utilized a technique that incorporates window slicing and window overlap, allowing the temporal network to process short, successive segments of the signal at any given moment. With this design, however, several trade-offs naturally arise due to the fixed nature of the window size. If the window is too short, the model may fail to recognize the gesture in time because the necessary change in signal amplitude may be spread across multiple windows. Conversely, selecting a longer window forces the model to retain a greater temporal span, which can make it more difficult to recall the most relevant features within that extended interval. These types of considerations are common when working with TCNs and are typically adjusted based on the research objectives.

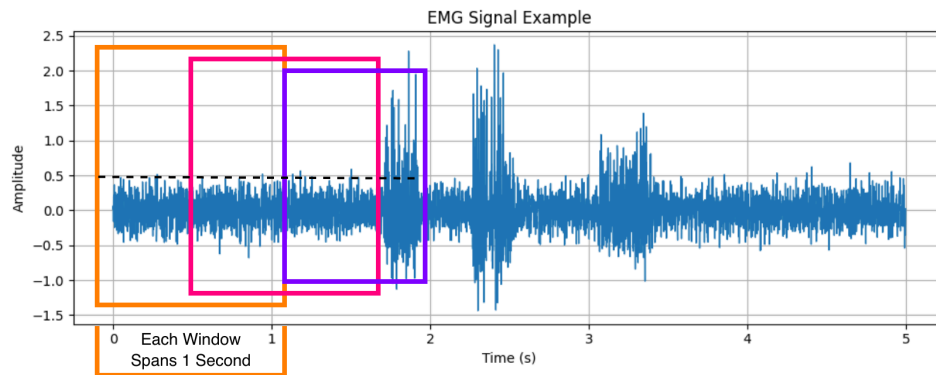


Figure [3]: Example of an EMG signal and how windows are partitioned.

The Figure 3 above demonstrates how we approached windowing in our neural network. Since each EMG signal spans approximately five seconds, a reactionary classification of a gesture should ideally occur within about one second. In practice, however, this is insufficient for real-time prosthetic control, where responsiveness to human motion is critical and delays are unacceptable. In such cases, reducing the window size accelerates gesture classification and enables faster prosthetic response. Another important component is the overlapping mechanism shown in Figure 3. By generating windows at one-second intervals but shifting them backward by 500 ms, we introduce the temporal continuity required for TCNs by allowing each window to interface with the raw EMG signal from the previous time step. This design enables the model to generalize patterns more effectively and retain information across sequential windows. The smaller the window gap, the stronger the temporal coupling and the more effective the model becomes at capturing the underlying dynamics of the signal.

Data Augmentation

Now that we understand the core concept of TCNs and how feature extraction is introduced at the window level, we can discuss the broader ambitions of this research paper. As

outlined previously, the goal is to augment the data extracted from the eight channels of the Myo Armband and utilize a model fusion strategy that aggregates these signals into additional descriptors, specifically the Mean Absolute Value (MAV) and the waveform length within each segmented window. While TCNs are designed to operate effectively on raw, naturally generated time-series data, our objective is to supplement the raw stream with features that preserve the essence of the original signal while enhancing temporal recognition. This motivates the inclusion of two additional attributes in conjunction with the raw inputs to the model.

The Mean Absolute Value serves a role analogous to a moving average for those familiar with stock trading indicators. Conceptually, it provides a means of reducing multi-channel EMG activity into a single value stream that captures the overall behavior of the summed channels averaged across each window. This offers the model a consolidated representation of how all channels evolve during a given segment. It is essential to emphasize that MAV is computed per window rather than across the entire signal. This distinction matters because calculating MAV across historical data, such as any segment preceding the 1-second threshold, would produce a mean influenced by earlier signal activity and therefore misrepresent the window's true characteristics.

Waveform length complements MAV by encoding how the amplitude of the EMG signal changes over time within the window. It provides the model with information about the signal's complexity, capturing total variation rather than average magnitude. By integrating waveform length alongside MAV, we introduce an additional layer of descriptive power that reflects temporal fluctuations within the signal. Together, these features supply the model with richer

insight into how each channel behaves in a given window, thereby supporting more informed temporal pattern recognition.

Model Architecture

We developed two variants of a TCN model: one that operates solely on the baseline stream of raw EMG values and another that incorporates data augmentation through the additional MAV and waveform length features. This model is designed to ingest windowed segments of the raw EMG signal. We intentionally kept the architectural complexity relatively constrained due to hardware limitations and the number of iterations required to refine the model effectively.

Breaking down the models further, we first established a baseline configuration to serve as a point of comparison for evaluating the effects of data augmentation. The initial design accepted only raw EMG inputs from a subset of 200 users selected from the total cohort of 612 participants. This subset produced approximately 60,000 unique EMG streams, each consisting of an eight-channel recording corresponding to a performed gesture. These streams were then passed into the baseline model, represented as the left architecture in Figure 4. Recall that what distinguishes a TCN from a conventional CNN in this context is the manner in which EMG signals are processed specifically through the windowing and overlap mechanism, which enables the network to capture temporal dependencies that CNNs are not inherently structured to retain.

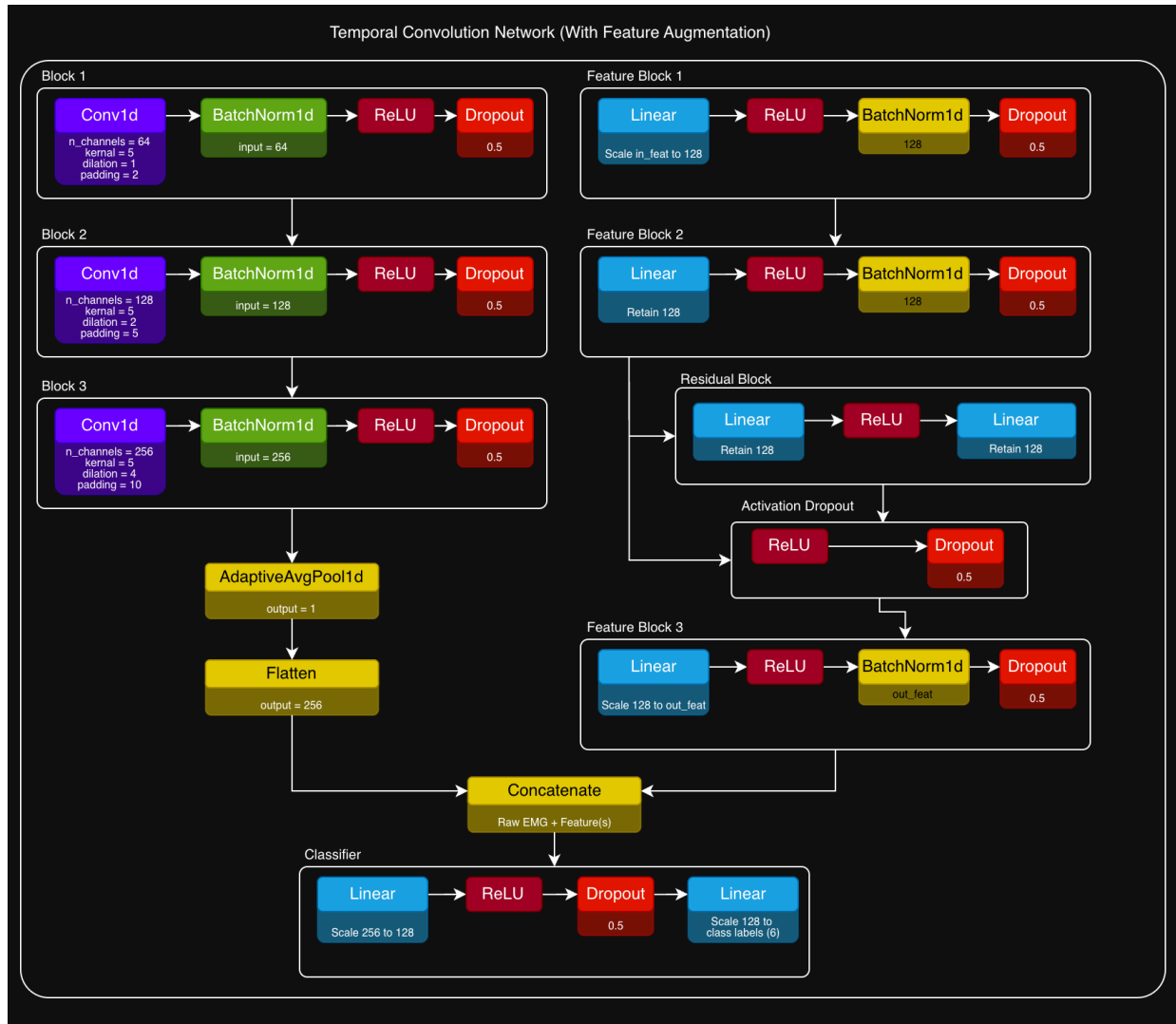


Figure [4]: TCN - left represents raw EMG input, right represents feature inputs (MAV + WL).

Expanding on Figure 4, the left model processes raw EMG inputs through a series of 1D convolutional layers (Block N) that increase in depth as the network propagates forward, consistent with traditional TCN dilation patterns. A critical component of this sub-model is the use of BatchNorm1d, which is particularly important given the variability in EMG signal amplitude, wavelength, and input size. Batch normalization ensures that the incoming data is

normalized across each batch, providing the model with a more stable and standardized foundation for learning. (Li, Yang, Peng, & Liu, 2021).

The dilated convolutional structure inherent in TCNs enables the receptive field to expand with depth, a behavior amplified by the windowing and overlapping scheme employed in our preprocessing pipeline. This design maintains temporal focus throughout training and allows the model to retain meaningful temporal patterns across windows. The inclusion of Adaptive Average Pooling serves to collapse the dilated temporal dimension into a compact, fixed-length representation before flattening, reducing computational burden while preserving essential information.

The right-hand sub-model (Features) functions as a streamlined Sequential Linear network augmented with a Residual Layer (GeeksforGeeks, n.d.; Shafiq & Gu, 2022). This component refines the integration of the MAV and waveform length features by learning how these manually computed attributes should align with the underlying EMG signal characteristics. Since MAV and WL are derived directly from the eight-channel EMG streams associated with each gesture, the residual layer helps stabilize their contribution and adjust their influence during training. Although the initial design omitted this residual layer, prior research indicated that such layers improve accuracy when the feature set carries meaningful discriminatory value, motivating its inclusion in the final architecture.

Materials, Devices & Data Sources

As previously discussed, the data source used to train this model is the EMG-EPN-612 dataset. This dataset contains recordings from 612 subjects, split evenly between training and testing groups. Each user contributes 300 EMG recordings, derived from 25 repetitions of each of the six gestures performed. Each recording spans approximately five seconds. In total, the full dataset includes 183,600 EMG samples; however, for the purposes of this study and the constraints outlined earlier, we selected a subset of 200 users from the training dataset, resulting in 60,000 EMG samples used for model development. The gestures included in our subset correspond to the provided labels: *Fist*, *Fingers Spread*, *Wave Out*, *Wave In*, *Double Tap*, and *Relax*.

MYO Armband

The device responsible for capturing this data is the MYO Armband, an eight-channel EMG sensor sampling at 200 Hz. The armband wraps around the user's forearm and records muscle activation patterns associated with the gestures being performed. These signals are processed by the device and then exported to a system that includes both the MYO Armband's internal gesture predictions and the raw EMG data, along with additional sensor outputs such as gyroscopic and rotational measurements. In this model, we focus exclusively on the



Figure [5]: Image from EMG-EPN-612 User (*Train, User1*).

eight-channel EMG signals; however, the additional measurements could be incorporated in future work to improve classification accuracy.

LibEMG (Open-source EMG Datasets)

The EMG-EPN-612 dataset was obtained from LibEMG (<https://www.libemg.com>), an open-source directory that hosts a wide range of EMG datasets collected globally, each tailored to specific use cases and measurement configurations. We selected this dataset in particular due to its comprehensive range of gestures and the characteristics of the recording device, which provides eight EMG channels sampled at a relatively modest rate of 200 Hz, an ideal configuration for our windowing and overlapping methodology. Another key advantage of this dataset is the repetition structure: each user performs every gesture 50 times. With 612 total participants (of which we used 200 for this study), the dataset offers substantial variability across subjects, enabling the model to generalize more effectively and learn from a diverse set of EMG patterns.

Data Extraction & Preprocessing

We initially acquired the dataset through the source: (<https://zenodo.org/records/4421500>) and subsequently constructed a custom DataLoader to iterate through and parse all files housed within the trainingJSON directory. This directory contains 306 subfolders, each denoted as userN, where N is an integer from 1 to 306. Each userN folder includes an image file and a corresponding userN.json file that stores the complete set of gesture recordings for that participant. As outlined previously, each user contributes a total of 300 EMG readings, each spanning approximately five seconds.

From each JSON object, we extracted the key information required for our study specifically, the "emg" and "gestureName" fields within the trainingSamples structure. The "emg" field contains a list representing the eight EMG channels, each expressed as a time-series of sampled values. The "gestureName" field provides the label corresponding to the gesture performed during that recording session. Once extracted, the samples for each channel are sorted and then stacked into an array with the following structure:

$$(n_{samples}, n_{channels})$$

Formula [1]: Shape of MYO Armband 8-Channel EMG readings.

With the raw EMG objects formed, the next step is window segmentation. Since the EMG recordings are continuous, they must be partitioned into fixed-length segments to ensure that each window is processed consistently. For this study, we selected a window size of 100 samples with a 50% stride (50 samples). The stride represents the degree of overlap between consecutive windows, as discussed earlier. Formula [2] below describes how each window is computed from the continuous signal.

$$window = \sum_{t=j}^n EMG[t : t + 100]$$

Formula [2]: Calculation of each window, **t = current timestep**,
j = previous end of time step and **n is next target timestep**.

The next stage of preprocessing involves handcrafted feature extraction for MAV and WL. For every sliding window, we compute these two features. To reiterate, MAV, Mean Absolute Value, is calculated as the average absolute amplitude of the EMG signal across all eight channels. Waveform Length (WL) is computed as the sum of the absolute differences between consecutive

EMG samples within the window. Together, these features provide complementary information: MAV captures the overall magnitude of muscle activation, while WL captures the complexity and variability of the signal over time.

Lastly, we used the LibEMG library in Python to extract a PCA analysis of the dataset for each gesture. Figure 6 below illustrates the resulting diversification of gesture-specific EMG signals following the standardization steps described previously. In the visualization, TR denotes the training examples and TE denotes the test examples. It is important to note that, while certain clusters are well separated, there remains noticeable overlap between several gesture groups. This overlap poses a challenge for the model, as it reduces the separability of the classes and makes it more difficult for the network to distinguish subtle variations between gestures. This motivates the introduction of manually computed features such as MAV and WL, which can help isolate clusters more effectively and enhance the model's ability to differentiate between gesture categories.

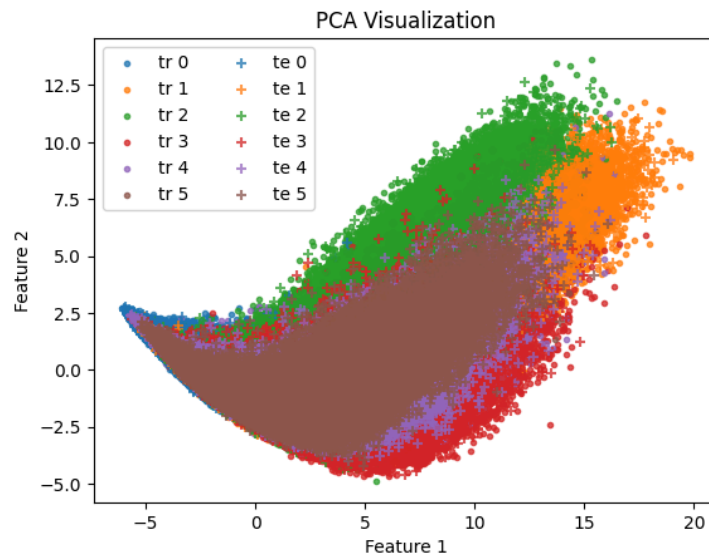


Figure [6]: PCA Analysis of 6 Gesture Labels.

Results

Based on the model architecture shown in Figure 4, we trained two distinct variants of the TCN model as previously described. The first served as a baseline and was trained exclusively on raw EMG data with no additional features. The second incorporated the augmented inputs MAV and waveform length to evaluate whether these features improved the network's ability to learn temporal patterns. Both models were trained for 30 epochs under identical hyperparameter settings to ensure a fair comparison.

Training was conducted on two separate offline hardware configurations: one utilizing an nVidia 3080 Ti GPU via CUDA acceleration, and the other using an Apple MacBook Pro equipped with an M3 Max chip running on Apple's Metal Performance Shaders (MPS). While the overall training results remained consistent across both systems, the dual-hardware setup enabled significantly faster experimentation, allowing us to iterate on model design choices and validate performance reproducibility across computing environments.

Figure 7 below presents the performance metrics specifically the training and validation accuracies and losses for both the baseline and augmented models. It is important to acknowledge that the loss seems to have more room to lessen and the accuracy to increase as per the slope of the last several epochs. Since we only trained it on 30 epochs to conserve time and resources, this would've allowed the model to mature and learn more from the data. These results form the foundation for the analysis that follows.

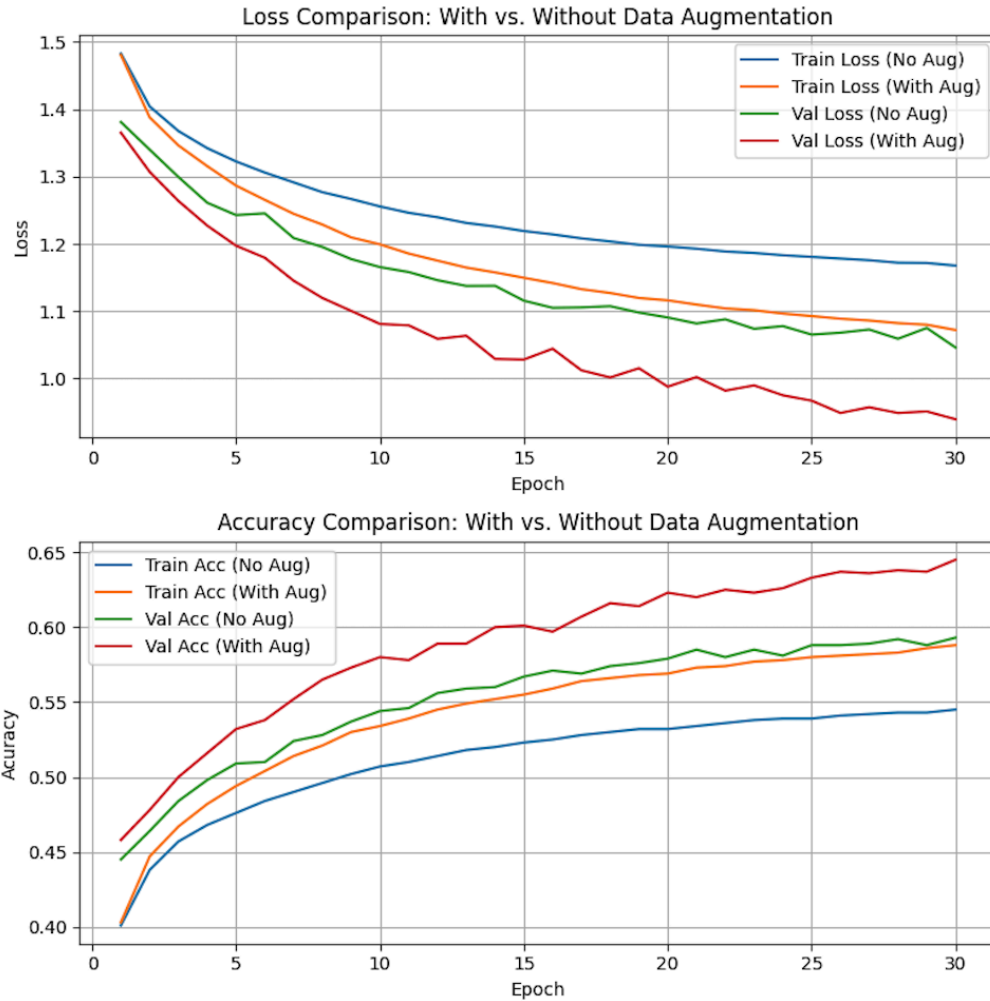


Figure [7]: TCN model loss vs. accuracy results (With vs. Without Data Augmentation)

As shown in Figure 7, the model incorporating data augmentation specifically MAV and waveform length demonstrated a clear improvement in performance. Accuracy increased from approximately 53% to roughly 60%, while validation accuracy improved from about 58% to nearly 65%. These gains indicate that the augmented model was able to leverage the additional feature streams to learn more effectively from the underlying temporal structure of the EMG signals. It is worth noting that these improvements were achieved after several iterations of model refinement, including the introduction of a residual component designed to enhance feature integration within the deeper layers of the network.

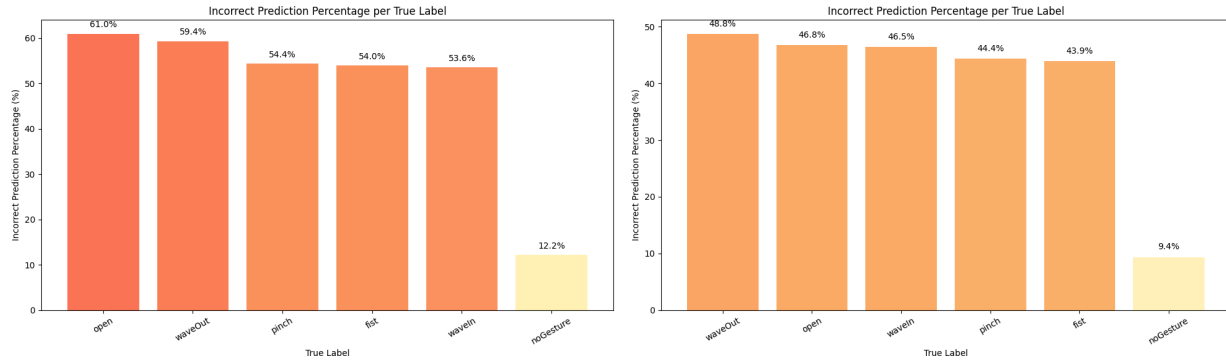


Figure [8]: Incorrect model predictions, left (no data augmentation) vs. right (with data augmentation)

Figure 8 presents a comparison of incorrect predictions across gesture classes for both model variants. The augmented model exhibits a clear reduction in misclassification rates, demonstrating its enhanced ability to distinguish between gesture patterns. The largest improvement occurred for the “open” gesture, where incorrect predictions decreased by 14.2%. The smallest improvement was observed for the “noGesture” class, at 2.8%. On average, the introduction of MAV and WL increased correct predictions by 9.13% across the gesture set. We hypothesize that the MAV feature in particular may have provided stronger discriminatory information for certain gestures, as it captures the overall activation magnitude across channels, enabling the model to identify distinct patterns within each window more effectively.

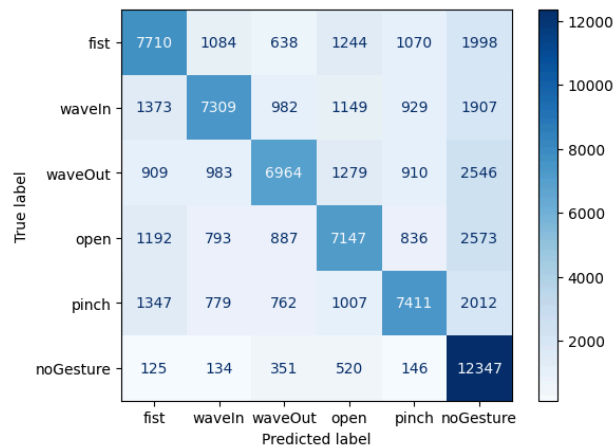


Figure [9]: Confusion matrix of the TCN + Data Augmentation model result.

Lastly, Figure 9 presents the confusion matrix for the TCN model with data augmentation, offering a deeper look into class-level performance. As indicated by the strong diagonal dominance, the model is generally successful in predicting the correct labels across most gesture categories. However, a noticeable amount of confusion persists between noGesture and the active gesture classes. This is likely due to the fact that we did not implement a sub-windowing method to isolate periods in which the user was actively performing a gesture. As a result, portions of idle signal were included in the dataset, and the model appears to have partially adapted to these non-gestural patterns.

Additionally, the confusion matrix highlights similarities in waveform characteristics between certain gesture pairs, particularly fist vs. pinch and waveOut vs. open. These overlapping signal patterns reduce the model's ability to reliably differentiate between these gestures. This suggests the need for expanded preprocessing techniques or additional feature extraction methods that can better isolate gesture-specific attributes. Incorporating more discriminative temporal or frequency-domain features, or refining temporal alignment strategies, may help separate these gesture domains more effectively in future iterations.

Discussion & Conclusions

As we have seen, augmenting the model with additional data features improved overall classification accuracy by an average of 9.13% across all gestures. Does this provide strong evidence that the right data augmentation strategies assist the model in learning? Yes, although, when compared to other work in the field, this merely scratches the surface of what TCN-based architectures have achieved on EMG signals. The core purpose of data augmentation, in this

context, is to enrich raw inputs with meaningful indicators that convey a broader picture at a faster and more interpretable level. For those familiar with stock trading, these features function similarly to technical indicators that encapsulate price, volume, and trend within a single metric. As alluded to earlier, this analogy was indeed an inspiration for the project: just as stock indicators summarize complex financial behavior, MAV and waveform length summarize essential characteristics of EMG activity so the model can learn more efficiently. This experience has opened the door to domains previously unexplored by the researcher and illustrates how impactful these models can be when applied to everyday human-centered applications.

There are, however, several considerations for future improvements both for the model itself and for manual data augmentation techniques. One major challenge lies in labeling and window alignment uncertainty. Physiological signals rarely exhibit strict, discrete boundaries, meaning that the exact start and end of a gesture often blur into surrounding activity. Since we sample only a snapshot in time, enforcing standardized and consistent window frames helps the model identify patterns more reliably. Cross-user variability is another important factor, as users differ significantly in wrist size, muscle strength, gesture consistency, and general muscular activation patterns. Addressing this variability may require larger datasets, domain adaptation methods, or more robust normalization techniques.

Regarding feature design, our study used only a simple set of MAV and waveform length. Future work could incorporate frequency-domain features, transient signal characteristics, amplitude dynamics, or more advanced temporal descriptors to provide richer information within each

window. The MYO Armband also records gyroscopic and orientation data, which we did not use but could substantially improve gesture discrimination in future models.

Thank you to the reader, we hope this research provides an encouraging perspective on how these technologies can be leveraged to improve human experiences and drive continued advancement in everyday assistive applications.

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Appendix

Use of AI-Assisted Tools

Portions of this research paper utilized AI-assisted tools for limited tasks, specifically for generating an illustrative image (Figure 1) and performing grammar refinement. These tools were not involved in the development of the models, data analysis, or any part of the experimental methodology. All conceptual design, implementation, interpretation of results, and written content were produced by the researcher.